Inertial Crosswalk-Crossing Detection for Visually Impaired Navigation Using Deep Learning

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***Abstract*—The advancement in portable and lightweight sensor technology made the inertial data from daily activities widely accessible, contributing to the interest of tracking these movements or detecting characteristic patterns in these movements. This study focuses on detecting the crosswalk-crossing behavior of visually impaired individuals, which are observed to differ from those of normal sighted people. We hypothesize that a tactile search of the crosswalk endpoint using a cane, a hesitation at the endpoints, or a particular steady walking while crossing are reflected as a unique series of readings in the inertial data. Crosswalk-crossings generate rare and relatively long-duration patterns that regular walking actions, making it challenging to determine which features of the sensory data to focus on. We propose a specially tuned learning framework for detecting crosswalk-crossings using Deep Convolutional Neural Networks, which are known to be effective at data analysis involving temporal hierarchies. Considering also the use cases of the detected crosswalk-crossings, such as correction in the pedestrian positioning systems, we emphasize a precision-focused training with weighted F-scores and a related loss function that minimize false detections. Our findings show that our approach successfully detects real-world crosswalk-crossings with high precision and acceptable recall values. We also show that the models are capable of performing similarly on the data from different devices they are not trained with.**

***Index Terms*—Convolutional neural networks, kinematic pattern detection, pedestrian navigation, visually impaired people, weighted F-score.**

I. Introduction

With the introduction of lightweight sensors into our lives, we have access to live *inertial data*, measurements of acceleration, rotation, and orientation of an object in space, captured by wearable sensors. These measurements provide real-time insights into movement, posture, navigation, and interaction with the environment, giving rise to the field of Human Activity Recognition (HAR). HAR generally focuses on classifying repetitive activities, such as standing, sitting, lying down, walking, running, cycling, and transportation [1], as well as daily actions like eating, drinking, and smoking [2]. Such data are typically processed in short time windows, enabling near real-time response while capturing meaningful patterns for most activities [3], [4]. By analyzing these patterns, HAR systems can yield valuable insights into human behavior, thereby improving safety, convenience, and personalization in everyday life, particularly for people with visual impairments.

Though closely related to and defined in the HAR domain, this study differs considerably from generic HAR approaches. First, we are interested in crosswalk-crossing behavior of people with visual impairments (PVI), which produces imbalanced data because such patterns happen rarely in normal walking conditions. Second, the patterns are longer than the conventional sliding window sizes (2-5 s) typically used in HAR. Despite the extensive HAR literature, position-relatable human behavior detection has received little attention, and currently no benchmark studies exist on crosswalk-crossing detection with inertial data.

PVI undergo specialized training to navigate safely and adapt to daily challenges for increased independence, learning skills such as crossing streets, using public transportation, and moving confidently in unfamiliar environments [5]. Building on this connection with HAR, our objective is to detect distinct motion signatures in inertial data from PVI that can be associated with specific positions on a map. Since orientation and mobility (O&M) instructors explicitly teach strategies for crossing streets and locating safe crossing points [5], PVI exhibit similar series of behaviors when they encounter a crosswalk. Thus we hypothesize that these behaviors manifest as recognizable patterns in inertial data.

Whereas PVI use various technology-enhanced systems, such as smartphones, smart glasses, or wearable cameras, to complement their impairments [6], traditional white canes or guide dogs remain indispensable due to their need for sensing the environment directly [7], financial constraints, usability challenges, and lack of alignment with user needs [8]. In addition, PVI prefer to keep at least one hand free, while the other is used for a cane or harness [9]. Studies [10], [11] and our own sessions with them [7] indicate that smartphones are typically carried in pockets for security, privacy, and natural mobility. We design our method for pocket-carry conditions, reflecting realistic deployment rather than laboratory usage. Practical constraints dictate that any approach must operate on widely available, affordable hardware without requiring additional devices or infrastructure, making in-pocket smartphones the most scalable platform. Accordingly, we prioritize sensing modalities that (i) are standard in smartphones, (ii) remain functional under pocket-

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carry, and (iii) are computationally feasible for real-time execution. In addition, inertial sensing provides a continuously available signal source that does not depend on external infrastructure or environmental visibility, which may limit the availability of GNSS or vision-based methods in real-world usage conditions. Inertial-based behavior detection meets these requirements, functioning without extra hardware or continuous satellite/camera availability. While this work focuses on crosswalk-crossings, it should be regarded as an initial step toward inertial behavior-based landmark detection. Pedestrians often exhibit consistent motion patterns when interacting with urban infrastructure, and detecting such patterns from inertial signals may enable their use as semantic correction cues in positioning systems. Crosswalk-crossing detection therefore serves as a first concrete demonstration of this concept.

We position the inertial crosswalk-crossing detection problem in the HAR domain, where reliably detected events can serve as correction cues for Pedestrian Dead Reckoning (PDR)-based positioning systems, which rely also on inertial data and are prone to drift due to accumulated integration errors [12]. Crosswalk-crossing detection is one such behavior-derived landmark event, which we formulate as a binary classification problem over inertial data windows: crosswalks correspond to positives and all other segments to negatives. A first challenge is that raw inertial data encompassing a crosswalk may have variable durations of several seconds, whereas feed-forward networks require fixed-size input. Observing that crossing a crosswalk takes at most 12 seconds, we fix the input window to 2400 samples at 200 Hz. Since this window is longer than those used in traditional HAR methods, standard features become smoothed and non-discriminative. We therefore use Deep Convolutional Neural Networks (DCNN), which have been shown to perform well on datasets with minority classes [13], and can capture inherent features of crosswalk-crossing patterns at multiple resolutions.

As the detected crosswalk information is to be used as corrector over a less reliable positioning system, their false detection rather increases the error than correcting the positioning. Whereas missing a crosswalk is acceptable, false positives are intolerable. Accordingly, another challenge of this research is detecting crosswalk-crossings with high precision and acceptable recall values. We attack this precision-bias by using a precision-weighted F-score metric, or namely F_β score, to satisfy this requirement by favoring precision over recall. Aligned with the metric, we also employ a related loss function, a combined surrogate precision loss function based on binary cross-entropy [14], to train precision-biased models.

This article is an extended version of the conference paper [15]. The first extension onto this work is the size of the data captured from the natural walks of the PVI, which is doubled compared to the previous version. Moreover, the data is enriched with data collected also from real urban areas and three more PVI. A second extension is the extracted datasets used in training. We have access to a naturally imbalanced dataset of inertial data, in which the crosswalk-crossings (positives) correspond to only about 3% of the whole data. To make the models better distinguish the crossings from the normal walking instances, we augment the data with small hop sizes of sliding windows, which leads to long training sessions. However, it is always likely that the original dataset may both under-represent the minority class (positives) and be extremely heavy and thus not scalable for training. Here, we analyze also the usage of undersampled versions of the majority class at two different levels, by generating reduced and balanced datasets along with the original dataset. A third extension is attained by testing the trained models on previously unseen smartphone data, which is considered to be more accessible for real-world conditions. Finally, we investigate further into the trained models through their responses to different parts of the input data, motivated by the five components of street crossing defined as a part of the O&M training [5]: arrival at curb, determining crossing point, alignment, analyzing traffic/pedestrian control and maintaining alignment. We provide an analysis of the important parts of an inertial data window.

This study makes several contributions. First, we formulate crosswalk-crossing detection for PVI as a rare-event HAR problem using inertial data from pocket-carried smartphones and long temporal windows that capture the full crossing behavior. Second, we propose a precision-oriented deep learning framework based on DCNN, trained with a precision-biased objective using the F_β metric and a compatible surrogate precision loss to minimize false positives. Third, we show the effectiveness of the proposed method on a real-world dataset collected from natural walks of PVI in urban environments, significantly increasing the amount of recorded data and preserving realistic smartphone pocket-carry conditions. Finally, we analyze the impact of class imbalance through different dataset configurations and evaluate the generalization of the trained models on previously unseen smartphone data while examining which parts of the inertial signals correspond to key crossing behaviors taught as a part of O&M training.

The next section, Section II, presents the state-of-the-art for landmark-based PDR correction studies. Section III, explains the precision-focused DCNN training methodology that specializes in crosswalk-crossing detection in inertial data. In Section IV, we describe the setup for the data collection and their preparation for training purposes. In Section V, we assess the performance of the trained models on the real-data, provide a discussion on the specific cases and comment on the limitations. We conclude our paper by summarizing our findings and provide future directions in Section VI.

II. Related Work

Whereas several studies have used structural landmarks to correct PDR drift, behavioral landmarks based on inertial data are hardly studied. For example, CRF-PDR uses key indoor positions such as stairs and elevators to build a pseudo indoor plan, identifies pedestrian activities at these landmarks, and estimates the optimal landmark sequence to correct PDR positions [16]. Similarly, a 3D PDR method matches motion patterns such as stair climbing and corridor turns to known structural features, calibrating accumulated errors in multi-floor buildings [17]. IMU-based detection of point-wise landmarks, including stairs, escalators, and corners, captures distinctive motion patterns with accelerometers, gyroscopes,

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and magnetometers. HAR-inspired methods such as LSTMs can recognize these landmarks with high accuracy, providing fixed reference points to improve PDR without extra hardware [18].

In terms of activity recognition, Alaoui et al. [19] use IMU, barometer, and GNSS data with a k-Nearest Neighbor classifier to detect walking, stair climbing, and cornering, identifying points of interest through activity sequences or transitions. Detecting position-wise points of interest, such as crosswalk-crossings, remains challenging, as they mainly consist of consecutive walking segments. Similarly, Khalili et al. [20] use accelerometer, gyroscope, and barometer data with SVM and Decision Tree classifiers to detect walking and stair activities, though both methods assume fixed device orientations, limiting applicability under real-world pocket-carry conditions.

In a previous study [21], we evaluate a fusion of PDR with different other modalities under pocket-carry conditions. The third-quartile positioning error of the fusion of PDR and GNSS is approximately 3.5 m. At pedestrian scale, this magnitude is non-negligible. In the same study, additional fusion of landmark-based corrections reduces the error below 2.5 m, indicating that additional non-GNSS cues can meaningfully improve localization accuracy under realistic usage conditions. Moreover, in a more recent paper, the detected crosswalk-crossing events detected by the proposed method are integrated into a real-time trajectory correction framework [22]. Corrections are applied at the moment a crosswalk-crossing event is detected (i.e., at the end of the inertial window), reflecting practical deployment conditions. The results demonstrate that crosswalk-crossing based correction reduces accumulated PDR drift by more than 50% in walking sessions of approximately 1 km, even with a limited number of crosswalk encounters. These findings provide quantitative evidence that reliably detected crosswalk-crossing events can serve as effective correction cues within a PDR framework.

Besides the behavior analysis on the crosswalks, we have access to a large body of literature on vision-based techniques, where studies detect specific types of crosswalk visuals (zebra, solid, standard, etc.) using smartphone cameras in real time [23], [24], or from aerial imagery [25]. These approaches employ modern neural network-based object detection methods, such as You Only Look Once (YOLO), to detect crosswalks and provide referential or absolute positions that can then be used to correct PDR. While these systems are sufficiently accurate, processing visual information on handheld devices is computationally expensive. Similarly, satellite positioning may be degraded in urban environments or totally unavailable indoors, for which beacon-based positioning systems can provide an alternative, though they require prior hardware installation and calibration [26]. Considering the preference of visually impaired users to carry smartphones in their pockets [7], [10], [11], vision- or GNSS-dependent approaches may be impractical for continuous real-world use. By contrast, maintaining both the PDR estimation and its correction cues within an inertial framework avoids these resource-intensive requirements, and allows the system to remain fully functional under pocket-carry conditions. This makes crosswalk-crossing detection a feasible, deployable, and practical solution for pedestrian-scale navigation, including environments with limited or no GNSS availability.

Despite these advances, most existing methods rely on fixed device orientations, vision-based inputs, or GNSS signals, which limits their applicability under realistic pocket-carry conditions and continuous daily use. Few approaches are based on inertial data alone to detect landmarks or behavioral cues robustly, and classical machine learning methods often struggle with long-duration or imbalanced activities such as crosswalk crossings. This underscores a need for advanced sensor-only solutions that can capture complex motion patterns directly from inertial signals. Our work addresses this gap by detecting crosswalk-crossing events from pocket-carried smartphones, providing a foundation for robust, deployable, behavior-based landmark recognition in real-world settings.

III. Methodology

We construct the problem of crosswalk detection through inertial data as a binary classification using convolutional neural networks. We use fixed-size windows of acceleration and angular velocity magnitudes as input to the classification system. Each instance is fixed to have two channels of 2400 samples, corresponding to 12 seconds at 200 Hz. In a realistic and natural walking scenario, an instance is processed to produce an estimation for the window that ends at that moment. If the person has passed over a crosswalk in that window, we expect to have an outcome of positive, or negative otherwise. Fig. 1 illustrates an example walking session with the magnitudes of three-axis accelerometer and gyroscope readings (Amag and Gmag), predictions, ground-truth crosswalks, and thresholded outcomes.

In order to train the networks, we assume to have access to a dataset labeled with positive instances that include crosswalk passages within and negative instances for all other situations that do not include a full crosswalk. In a realistic dataset as the positive instances happen to be very few compared to the negative instances, the set of negative instances and the set of positive instances will also be called as the majority class set and the minority class set respectively.

The classification problem is handled differently than a conventional classification problem. We follow a precision-focused training and evaluation technique, that is, false crosswalk detections are forced to minimum, whereas missing the crosswalks is tolerable to a degree. We first explain how this focus of precision is achieved. Secondly, we show how to customize the convolutional neural networks for detecting crosswalks.

A. Precision-Focused Accuracy Scores and Loss Function

In a traditional binary classification, the aim is to optimize a function that yields the best general accuracy, which is achieved through a balance between false positives or type I errors, and false negatives or type II errors. However, our motivation is not only to detect the crosswalk-crossings, but also use this information as a corrector in a prospective navigation algorithm. Incorrect detections or false positives would

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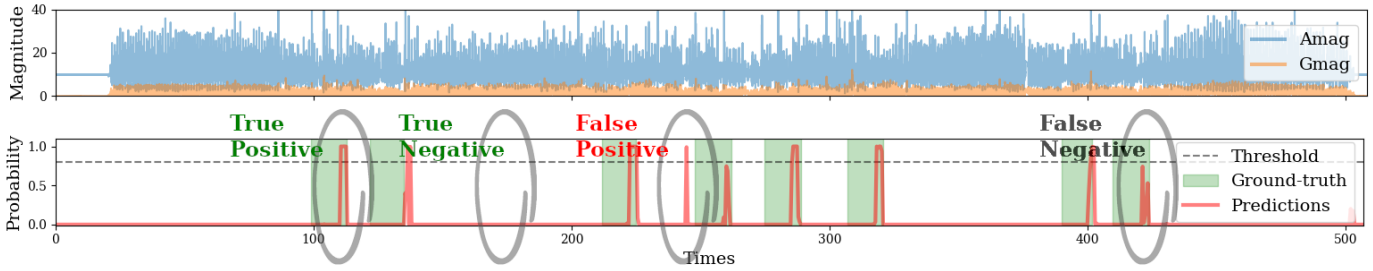


Fig. 1. A visualization of a walking session in terms of magnitudes of accelerometer and gyroscope readings (Amag and Gmag) (top); and possible outcomes after thresholding the prediction probabilities along with true crosswalk-crossings (bottom).

disrupt such a correction operation. Contrary to the traditional approach, where the decision threshold is set at 0.5, we instead use a much higher threshold (>0.8), as illustrated in Fig. 1. In this figure, the green patches indicate the true crossings, while the red line represents the predicted probability of crossings. Our method requires a high level of certainty before classifying an event as a crossing.

We shift this certainty requirement also to the training phase to train models that are sensitive to false positives (FP). We accept only extremely successful predictions as positive values, which may lead to classifying a high number of false negatives (FN), by ensuring a lower number of FP. We employ two well-known metrics to measure the performance: Precision and Recall. The former is for identifying true positives (TP) among all predicted positives and the latter for identifying TP among all actual positives:

$$\begin{aligned} \text{Precision} &= \text{TP}/(\text{TP}+\text{FP}) \\ \text{Recall} &= \text{TP}/(\text{TP}+\text{FN}) \end{aligned} \tag{1}$$

Our objective corresponds to prioritizing precision over recall. However, totally neglecting recall is also undesirable, since a model can trivially achieve high precision, up to 1.0, by predicting very few TP with no FP. To prevent this situation, we require recall to remain at moderate levels, though not necessarily close to 1.0. This trade-off is quantified by replacing the traditional F-score with a more generic F-beta score, F_β [27], defined as:

$$F_\beta = (1+\beta^2)\frac{\text{Precision}\cdot\text{Recall}}{\beta^2\text{Precision}+\text{Recall}} \tag{2}$$

where β is the focus or importance parameter, for higher values ($\beta > 1$) of which recall is prioritized over precision. For our goal of giving priority to the precision, we set $\beta < 1$.

In the training process of a binary classification model, we employ a combined surrogate precision loss function [14] as follows:

$$\mathcal{L}_\beta = \overbrace{-\frac{1}{N}\sum_{i=1}^{N}\Big[y_i\log(\hat{y}_i)+(1-y_i)\log(1-\hat{y}_i)\Big]}^{\text{Binary Cross-Entropy (BCE)}} + \underbrace{\beta\left(1-\frac{\sum_{i=1}^{N} y_i\,\sigma\big(\alpha(\hat{y}_i-\theta)\big)}{\sum_{i=1}^{N}\sigma\big(\alpha(\hat{y}_i-\theta)\big)+\varepsilon}\right)}_{\text{Surrogate Precision Penalty (SPP)}} \tag{3}$$

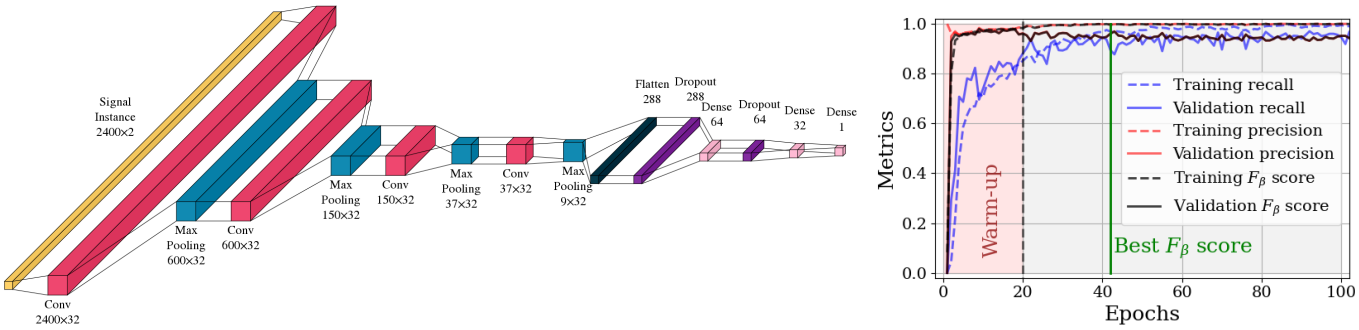
which is merely an addition of the well-known binary cross-entropy (BCE) and a surrogate precision penalty (SPP) that favors the precision with respect to the precision focus parameter, β. Here, $\hat{y}_i$ and y_i correspond to the estimated and true labels of the instances respectively, N is the total number of instances in the training batch, i are the instance indices, and ε stands for a small number to avoid division by zero. To make SPP differentiable, the decision is taken through with a steep sigmoid function, $\sigma()$, with the help of the threshold parameter, θ, and the steepness parameter, α. The values of the precision focus, β and the decision threshold, θ, are kept fixed both in the training phase with the proposed loss function and in the prediction or performance measurement phases.

B. Deep Convolutional Neural Networks for Crosswalk-Crossing Detection

We employ DCNNs that are constructed as a series of several convolutional and pooling layers, a flatten layer and a dense Multi-Layer Perceptron (MLP), as illustrated in Fig. 2a. Considering the long durations of the motion of interest, the intuition behind the usage of deep convolutional layers is to keep the relation between large portions of input data while downsampling the input signal gradually in a pyramidal way to capture the underlying structures. The depth of the convolutional layers will be denoted by c. With $c = 2$, we have a classical Convolutional Neural Network (CNN) that has two convolutional layers, each of which is followed by a max pooling layer. The CNN architecture is constructed in a traditional way, however, the parameters are tuned unconventionally. As an example, the kernel mask size of the convolutional layers is set to be $k = 15$.

We also fix the number of features at the flattened level by adapting the number of pooling sizes at max pooling layers after each convolutional layer. The architecture of the dense layers is also fixed in order to investigate the impact of selecting a different number of convolutional layers. In the dense layers, the activation function is selected as the Rectified Linear Unit (ReLU), with its effectiveness in mitigating the vanishing gradient problem and its ability to promote sparsity and support efficient gradient propagation, which significantly enhances the training of deep networks. The sizes of units are set to 64, 32, and 1, for the last of which the activation function is naturally *sigmoid*, as we need a single final estimate. We choose a moderate dropout probability of 0.1, to add artificial disturbances into the training process. As another technique to favor precision in the training process, we tune the decision threshold of the classes, θ, to higher values than the traditional

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(a) An illustration of the DCNN with 4 convolutional layers ($c = 4$). (b) Evolution of scores in a typical training trial.

Fig. 2. DCCN model and training.



(a) Experimental track and its crosswalks (b) Urban area and its crosswalks (c) Capture devices (d) Device placements

Fig. 3. Data collection setup.

value of 0.5, by which we aim to decrease the loss and increase the weighted F-score at the same time.

The other tunable parameters of the training play a minor role in this aim. We optimize the models with the Adaptive Moment Estimation (Adam) method [28], as it combines the benefits of momentum and adaptive learning rates, leading to faster convergence and better stability. We select a batch size of 128, which provides a balance between training stability and computational efficiency. We observe that smaller batch sizes introduce more noise in the gradient estimates, slowing the convergence, while larger batch sizes require more memory and reduce the models ability to generalize. The training is run up to 200 epochs to ensure that the models have sufficient opportunity to converge. Among these epochs, the weights that produce the best weighted F-score on the validation set are captured as the best model weights, whatever the final performance or the loss are. Moreover, we observe that the best model in a training session can be acquired before the 100th epoch in most of the cases as was given in the example of the training procedure Fig. 2b.

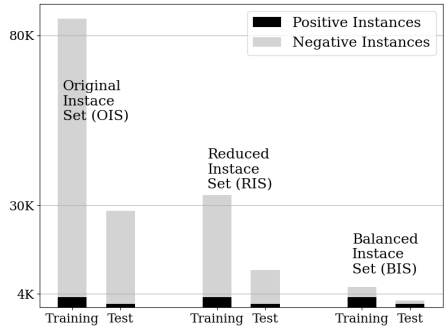
IV. Experimental Setup

A. Data Collection

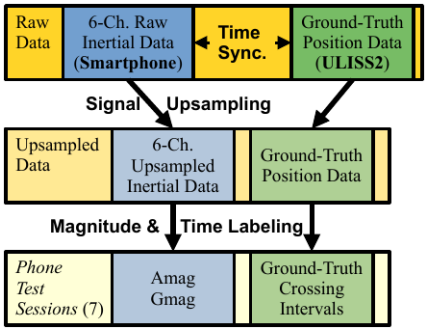
The inertial data that make up the skeleton of this work was collected as a part of another study that investigates the habits of PVI with a focus on personalized PDR [29]. Here we use data from seven different PVI with ages between 45 and 60, three of whom are women and four are men, with heights ranging from 157 to 185 centimeters. The level of visual impairment varies across participants (5,3,2,2,1,5,5), with five corresponding to complete blindness [30]. Although they have different walking styles [29], none of the participants showed signs of walking impairment or any other health issues during data collection. Moreover, the data were collected while they walked using their ordinary white cane, without any other assistive tools.

With the informed and signed consent of each participant, they are asked to walk along two different tracks. The first one is an experimental track constructed in the campus that has multiple crosswalks of different dimensions and orientations, as well as the other urban objects used by the participants reference points, such as audible traffic lights, post boxes and pavements (see Fig. 3a). Whereas this track poses less risk for data collection, our participants remark that it provides few environmental cues, given the low traffic and limited building structured that typically create the wall effect they rely on in urban areas. Moreover, unlike real urban conditions, the sidewalks contain grass and small stones which may influence their walking behavior. The second environment is a real urban area that includes all the city-related obstacles that PVI may face (see Fig. 3b). The streets of both environments are open to vehicle and pedestrian traffic. Whereas the crosswalks in the

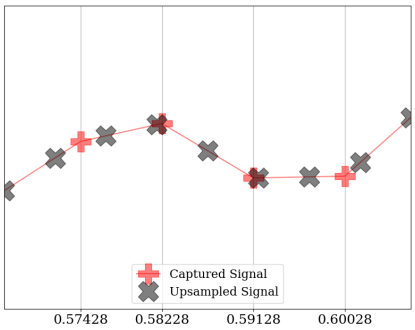
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(a) Distribution of the instance classes used in training and and model assessment



(b) Dataset preparation for real-world evaluation



(c) Upsampling the original signal to the DCNN-compatible input

Fig. 4. Dataset preparation.

experimental track are newly paint and properly structured, the ones in the urban area may have skewed forms or deformed tactile bands at their endpoints, both of which may affect the walking behavior of the participants. As also seen in Fig. 3b (bottom right), the hemispherical bollards at the endpoints of crosswalks create risks for the PVI and cause different reactions in such situations. We note that the data also includes passages over consecutive crosswalks that are used to cross divided streets (see Fig. 3b - bottom left).

For this study, we primarily select zebra (continental) crossings so that drivers can easily recognize pedestrian crossings and thereby reduce security risks during data collection. However, we believe that the type of the crossings would not affect the behavior of crossing the street, whether they are standard, solid, dashed, etc, since this study does not use any visual data captured from the environments, and more importantly the participants report that their experiences while crossing crosswalks are similar in all conditions, likely involving hesitations and brisk search behaviors at the terminals of the crosswalks.

We use two different devices to collect inertial data. The first one is the Ubiquitous Localization with Inertial Sensors and Satellites (ULISS) device [31], which is an advanced inertial navigation system that includes an Xsens Mit-7 IMU-Mag sensor and a GNSS receiver (Fig. 3c top). The device provides triaxial acceleration, gyroscope and magnetometer readings at 200 Hz, synchronized with GNSS timestamps for accurate corrections. The second one is a Google Pixel 6 with a custom application that reads inertial data at about 100 Hz and records them in the same format (see Fig. 3c bottom).

ULISS or the smartphone are placed in either a torso pocket or a trouser pocket of the participant as shown in Fig. 3d. Aligned also with the data collection for personalized dead-reckoning purposes [29], we do not consider a fixed orientation, however the devices are put in a body-stable manner to avoid possible bouncing movements that would introduce extra noise in the inertial data. Participants are instructed to walk freely on predetermined tracks ranging from 700 m to 1200 m, with minimal guidance. Because the tracks may be new to the participants, some emergency guidance is provided to them in risky cases of tripping due to an unperceived obstacle. Their walks are recorded in terms of inertial data, namely accelerometer and gyroscope readings. Alongside the inertial data, through another ULISS placed on the head of the participant, we capture GNSS positions that serve as the ground-truth for positioning and labeling needs. The horizontal and vertical position accuracies of this secondary device on the head are reported as 1.5 m and 2.0 m respectively [32].

B. Data Labeling and Extraction for Training

We use 6-channel inertial data from one ULISS and true positions from the secondary ULISS. The two data are synchronized using GNSS time. As we expect to have variant orientations of the devices, the data are processed into their magnitudes. For the sake of consistency, we will use the term *sample* for a single momentary data point of these data, *instance* for a vector of 2400 samples (12 seconds) in two channels, and a *walking session* for a whole series of two-channel inertial data. Our dataset include 59 walking sessions, of which we choose 10 full walking sessions randomly (at least one from each participant), and call them the *test sessions*. The remaining 49 sessions are used in the training, naming them as the *training sessions* (see Fig. 5). The timestamps of the crosswalks on all of the sessions are labeled by hand, after being verified on the true positions. As labels, we note the beginning time of the crosswalk and its duration.

We slide fixed-sized windows of 12 seconds on the training sessions, with a hop size of 0.25 seconds, corresponding to an overlap of 0.98, and save these windows as separate instances. The instances that encompass a full crosswalk-crossing within are labeled as positive instances. All the rest are labeled as negative. Note that a negative instance may also include portions of crosswalk-related inertial data. We get the set of extracted instances, making 4000 positive instances versus 120K negative instances. We call this set it the *original instance set* (OIS). Considering the training speed and further scalability of this study, we generate a *reduced instance set* (RIS) by uniformly sampling negative instances to form a set that is 10 times the size of the positive set. Finally we form a *balanced instance set* (BIS) by undersampling the set of negative instances as many as the positive instances. We shuffle and divide the final instance sets into training and test sets with proportions of 75% and 25% respectively to let a model learn

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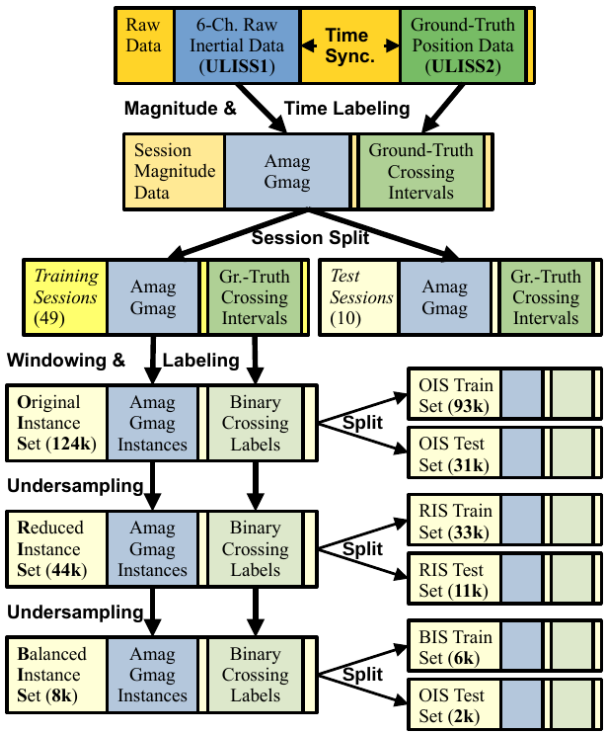


Fig. 5. Dataset preparation for training.

with a large number of instances. After the division, the test sets are used to assess the performance of the trained models. A visual comparison of the datasets that are used in training is given in Fig. 4a.

C. Data Extraction for Real-World Evaluation

We also have access to 7 walking sessions recorded with the smartphone, namely the *phone test sessions*, all of which are used for real-world evaluation. These session data are also exposed to the same magnitude operation to get orientation-free inertial data as shown in Fig. 4b.

However, before all, the data samples are upsampled to 200 Hz to make them compatible with the input shape of the DCNN models. The upsampling is performed by a linear interpolation of the original signal at equally spaced points corresponding to the required frequency as shown in Fig. 4c.

V. Experiments and Results

A. Experiments

The models are trained on an Ubuntu-based mobile computer equipped with an Intel Central Processing Unit (CPU) model of i7-6820HQ at 2.7 GHz and a Random Access Memory (RAM) of 16 GB. We also make use of the integrated NVIDIA Quadro M1200 Graphics Processing Unit (GPU) featuring 640 Compute Unified Device Architecture (CUDA) cores and 4 GB of RAM. The versions of the Nvidia Driver and CUDA Toolkit are 550.120 and 12.4 respectively. The training trials are performed using Python v3.11 and Keras v.2.7.0 under Tensorflow 2.16.1. At each trial, an instance set is divided into training and validation subsets randomly with a proportion of 75% in favor of training.

TABLE I

Comparison Table of the CNN Models With Respect to Their Max Pooling Level Configurations, Number of Trainable Variables, Average Training Epoch Duration by Using Different Datasets and Their Prediction Durations per Instance

c	Max Pooling Configuration	Size of Variables	Avg. Epoch Duration (s)			Predict Dur.(ms)
			OIS	RIS	BIS	
1	256	21601	20.69	8.09	1.55	0.86
2	16,16	36993	21.59	8.39	1.60	0.87
3	4,8,8	52385	29.79	11.56	2.19	0.92
4	4,4,4,4	67777	31.08	12.09	2.29	0.93
5	2,2,4,4,4	83169	55.16	21.38	3.49	0.97
6	2,2,2,2,4,4	98561	55.85	22.90	3.81	1.00
7	2,2,2,2,2,2,4	113953	57.08	23.58	3.87	1.04
8	2,2,2,2,2,2,2,2	129345	57.66	23.81	3.91	1.07

In a typical training process, it is observed that precision may misleadingly jump to 1.0 during the initial epochs, while recall remains at 0.0. This corresponds to situations where there are very few true positives and no false positives. These outcomes indicate that the model is far from being generalizable. To avoid such situations, a warm-up period of 20 epochs is enforced at the beginning of a training trial, as seen in Fig. 2b. After the warm-up period, the model of the epoch that achieves the highest F_β value in the validation set is saved as the best model for the current parameter configuration. As recall increases, F_β is observed to fluctuate, because of the enforced disturbances by dropouts, which allow the training process to occasionally find better and generalizable scores.

B. Performance on the Instance Sets

We first investigate the performance of the models on separate randomized instance sets that are not used in any training or validation processes. We analyze the influence of the depth of different CNN models, which is increased up to 8 layers. This design is achieved by selecting pool sizes in a balanced way such that their product is always 256 and repeated pooling operations does not result in collapsing to zero length. Because they should be integers, a perfect balance is not possible every time. In such cases, higher factors are put into the later layers, by which we attain the pool size configurations in TABLE I, where we also report the number of trained variables, the duration of each epoch in training with the instance sets, OIS, RIS and BIS and the duration of prediction of a single instance.

The training process is repeated to obtain models with varying network depths with different configurations of $\beta \in [0.05, 0.2]$ and $\theta \in [0.8, 0.95]$, the ranges that consistently strongly favor precision over recall. Considering the F_β-score (2), we note that the interval for the β parameter states that precision is effectively 25 times to 400 times more important than recall, or skipping non-existent crosswalks is more crucial than detecting existent crosswalks, for the sake of PDR correction. These intervals are chosen empirically but in line with our motivation to encourage the models to minimize the number of false positives. However we do not consider $\beta = 0.0$ or $\theta = 1.0$, so as to preserve room for improving recall by reducing false negatives too.

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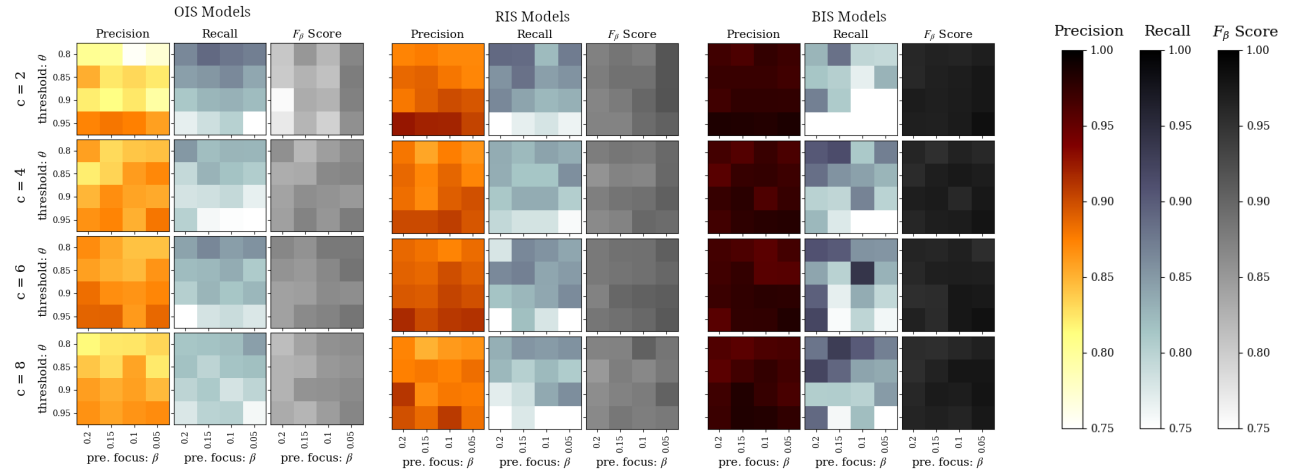
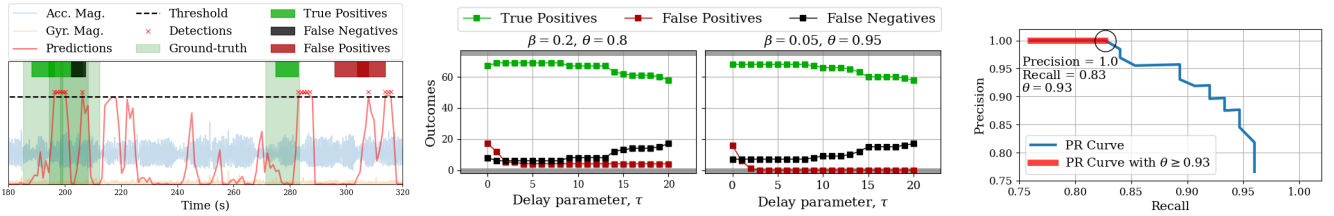


Fig. 6. Classification scores on the instance test sets.



(a) A sample from walking sessions with different cases of detection outcomes.

(b) Classification outcomes (TP, FP and FN) of the test sessions with varying delay values.

(c) PR curve of the model with $c = 4$, $\beta = 0.1$ and $\theta = 0.95$

Fig. 7. Comparison and evaluation of test outcomes.

In Fig. 6, we report precision, recall and F_β score on the test splits of OIS, RIS and BIS. Note that the scores are computed with respect to the ground-truth binary labels of the instances and presented in the interval of $[0.75, 1.0]$. We observe that the models trained with OIS struggle to pass the level of 0.8 for the F_β score, which is far from being sufficient. The same problem applies for RIS with the level of 0.95 for the F_β score, which is not compatible with our future PDR correction requirements. Moreover the models that have high precision scores (dark cells) fail to have acceptable recall scores (white cells), which refer to too many false negatives. Generally, training the models with BIS leads to obtaining models that perform well on the test sets with some acceptable recall scores. While a model with two to four convolutional layers performs generally well for different parameters, the contrasting differences in the deeper levels can be interpreted as that the models work either perfectly or lose generalization to the test set. We also observe that after two layers the models display similar performances, which makes the models with four layers more appealing as there is a processing performance degradation as of the fifth layer, as depicted in TABLE I.

According to these results, we can infer that F_β scores above 0.99 can be achieved for specific parameter configurations especially with the BIS, however, the models will highly likely lack information about different walking situations in terms of diversity because they are confined in the undersampled sets. Thus, these results may be misleading and may not cover the real-world diversity.

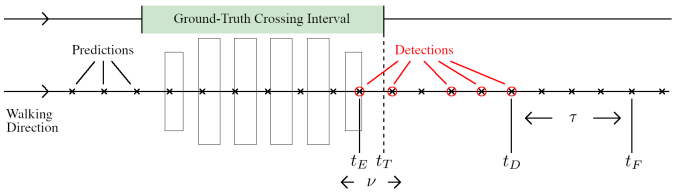


Fig. 8. Delayed decision and ground-truth matching technique around a crosswalk-crossing detection.

C. Performance on the Test Sessions Set

A more realistic batch of tests are performed on the test sessions set. The inertial magnitudes related to a session are again segmented into overlapping 12-second-long sliding windows with a hop size of one second. Each window is fed to the selected DCNN model to generate a binary prediction. These test sessions include 75 crosswalk-crossings, labeled as ground-truth crossing intervals in Fig. 5. As we are interested in detecting the crosswalk-crossings rather than classifying isolated instances, we measure the performance by the capability of matching the true crosswalks.

An illustration of detections around a crosswalk is given in Fig. 8, in which a participant is walking from left to right on the black line. A trained model is run at each second to generate a binary prediction, marked with "×". As the length of an instance is usually longer than the time to cross a crosswalk, we receive a series of detections as from the exit of a crosswalk, marked by "◯". These series of detections are likely to have gaps of missed detections (false negatives). To

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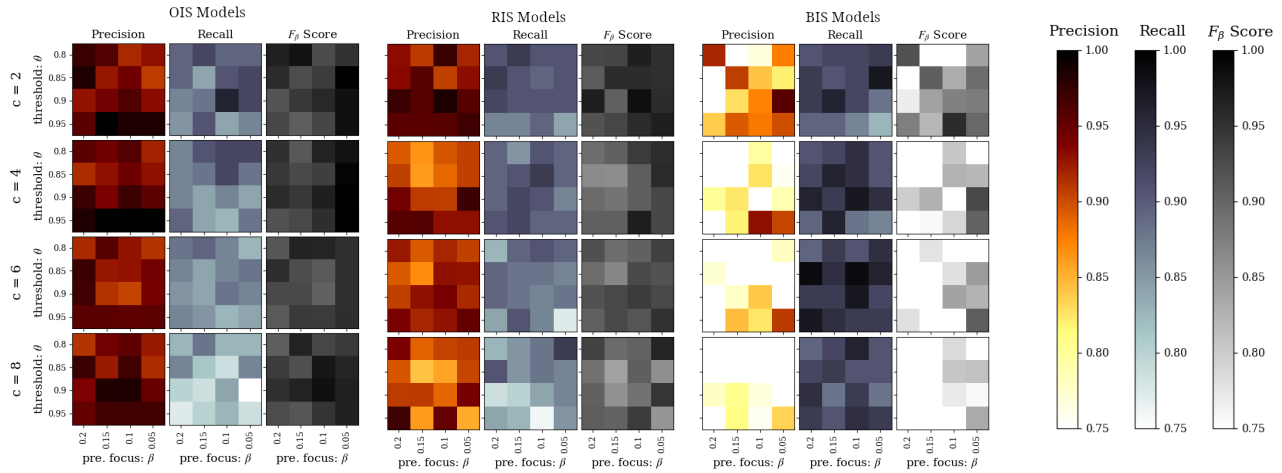


Fig. 9. Classification scores on the session test set by models trained with OIS, RIS and BIS.

avoid acquiring multiple series for one crosswalk, we employ a delayed decision mechanism that enforces a delay interval, τ, triggered by a detection. Whilst the models are designed to miss doubtful instances, the mechanism effectively yields a crosswalk-crossing behavior by grouping the detections and omitting the intermediate misses.

The delay interval is over at time t_F without any more detections within. Here, we can associate the beginning of the detection series, t_E, with the label of the time related to the end of the crosswalk, t_T. If the time difference between these two times is smaller than a neighborhood parameter, $|t_E - t_T| < \nu$, then we set this match as a true positive. If a detection cannot be associated with a true crosswalk, then it is a false positive. If there are true crosswalks that cannot be associated with any detection, then they are marked as false negatives. In this work, we fix the neighborhood parameter to $\nu = 5$ seconds. A prediction example on raw data along with the matched detections and the associated classification outcomes are presented in Fig. 7a. Note that this example is intentionally generated with a very inaccurate model to include the cases of false positives and false negatives. Two of the three consecutive crosswalk-crossings in the beginning are missed (black) and two crosswalk-crossings (red) are falsely detected towards the end.

After fixing the neighborhood parameter, we vary the delay parameter, τ, between 1 and 25 seconds to investigate its influence on the classification outcomes. To obtain the results given in Fig. 7b, we choose CNN models with $c = 4$ layers trained on OIS. We also take variations of other classification parameters, β and θ, into account. We observe that a delay of at least three seconds is mandatory. Below this value, the detections are not sufficiently filtered and yield extremely high false positives. As shown in the specific case of $\beta = 0.05$ and $\theta = 0.95$, we can even achieve no false positives with a low number of false negatives, which satisfies our target case of minimizing FP without losing too many TP to FNs. The delay values between 3 to 7 seconds can be accepted as safe values. Thus, we choose a delay value of 5 in further experiments, which can be interpreted as that the detection happens 5 seconds after the actual crossing is completed.

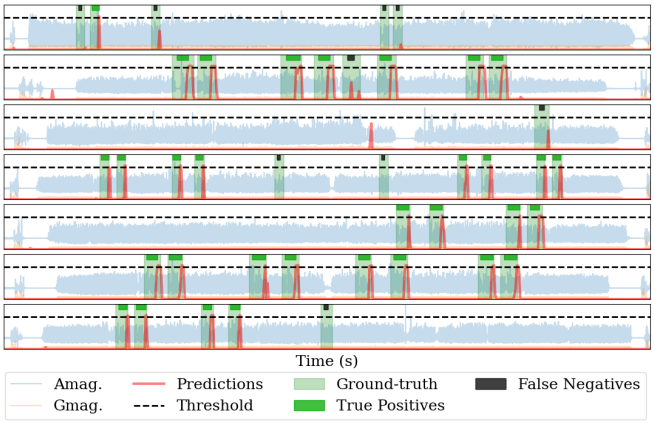


Fig. 10. Predictions and outcomes overlaid on the inertial data of the smartphone sessions set.

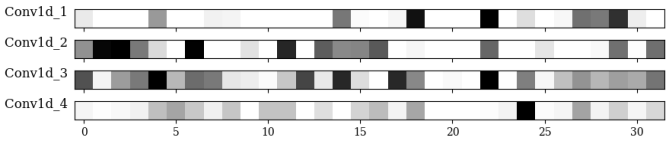


Fig. 11. Visualization of the importance of the separate kernels in the convolutional layers.

We report the performance of the previously trained models of the session test set in Fig. 9. Contrary to the results of the instance test sets, we realize that models that are trained on OIS perform better than its undersampled versions. This is an expected consequence, as the models have much more diverse knowledge about normal walking conditions and they can distinguish the crosswalk-crossings better. An advantageous finding here is that the models that are trained with RIS tend to perform close to the models trained with OIS, by which faster training is possible without any important expense in the detection performance.

We investigate the performance of the selected binary classification model through its Precision-Recall (PR) curve by varying the threshold parameter at different levels. We fix the

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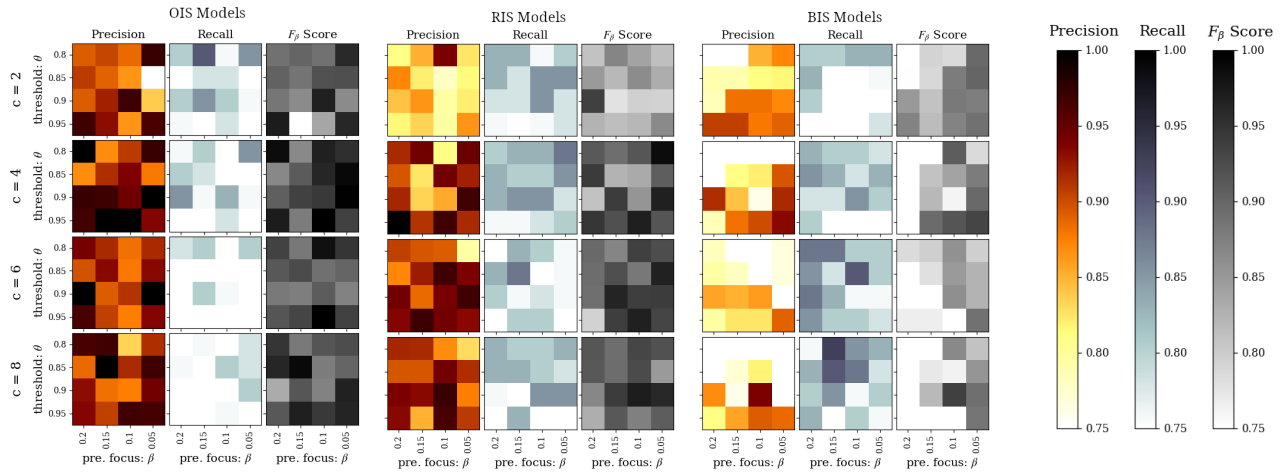


Fig. 12. Classification scores on the smartphone session test set by models trained with OIS, RIS and BIS.

model with parameters $c = 4$, $\beta = 0.1$ and $\theta = 0.95$ and attain the PR curve given in Fig. 7c, where the strong bias toward precision is easily distinguishable. The model performs best at higher thresholds ($\theta \in [0.9, 0.96]$), and precision is perfect for $\theta \geq 0.93$. Considering that false positives are very costly for our application, we choose the point of $\theta = 0.93$, yielding $F_{\beta=0.1} = 0.998$ with no false detections and 16 misdetections, which is an acceptable score. The PR Area Under Curve (PR-AUC) score is calculated as 0.95.

D. Performance on the Smartphone Sessions Set

We measure the performance of the trained models in the previous sections on the inertial data captured by the smartphone, without retraining. The results follow a similar tendency with the results of the test sessions set, except for the results of the balanced instance models, which are far from serving our goals as shown in Fig. 12. Whereas the models trained with OIS or RIS do not perform generally as accurately as the previous results, we attain occasionally some parameter values such as $c = 4$, $\beta = 0.1$ and $\theta = 0.95$, which yield Precision = 1.0, Recall = 0.781 and $F_\beta = 0.997$. The predictions and the outcomes of this specific model throughout a session is displayed in Fig. 10. The model misses a number of crosswalks, shown as the black bars, but it does not provide any false detections. Therefore such a model will provide a positioning algorithm with reliable crosswalk-crossing detections without misleading it with false detections.

Even though the models are not trained with the smartphone data, they can respond to the needs without any retraining or fine-tuning with the smartphone sensor data. Therefore we deduce that the models learn the general inertial patterns of a crosswalk-crossing rather than sensor-related patterns.

E. Feature and Input Analysis

Use of CNNs allows us also to interpret the models in terms of features, as convolutional layers produce feature maps in the intermediate steps [33]. In Fig. 11, we show the influence of the kernels of each layer on the final prediction. We immediately see that some of the kernels stand out by having high importance values (dark cells) in different layers. By feeding the models with diverse data, we see consistently the same pattern in the kernel importance figures, which is interpreted as that the same kernels happen to be the most important ones for the whole dataset. We plot the kernels that have the highest importance values in the first layer in Fig. 13 with respect to their channels. We see that the kernels tend to form similarly. The kernels 18 and 29 have similar forms in the accelerometer channel, and the kernels 22 and 29 have similar forms in the gyroscope channel.

We also investigate which temporal part of the input windows contribute most to the predictions. To attain this information, we employ the Gradient-Weighted Class Activation (Grad-CAM) method [34], in which we first extract the feature maps of the last convolutional layer as it includes high level spatial features. These features are then used to calculate the gradients of the target class score with respect to the feature maps. Gradients are averaged across spatial dimensions to compute the importance of each feature map. We attain the importance heatmaps by weighing the feature maps with their importance scores by summing and passing them through a ReLU filter. Finally, the constructed heatmaps highlight the influence of different time intervals of the input signal on the prediction. The importance heatmap in Fig. 14 is obtained by applying the mentioned method for the selected four-layered CNN model on the session set. We see that the beginning and the end of an instance are the most influential parts of the input for the prediction.

F. Discussion

We also propose a possible real-time implementation of the technique into a PDR-based positioning system. At time t_F, two important timestamps can be identified: the beginning and the end of the detection series, t_D and t_E, respectively (see Fig. 8). The latter can be associated with the end of a crosswalk, while the former can be linked to the beginning of a crosswalk by subtracting 12 seconds, yielding $t_S = t_D - 12$. Furthermore, the midpoint $t_M = (t_S + t_E)/2$ is expected to serve as a reliable indicator of the center of a crosswalk. With

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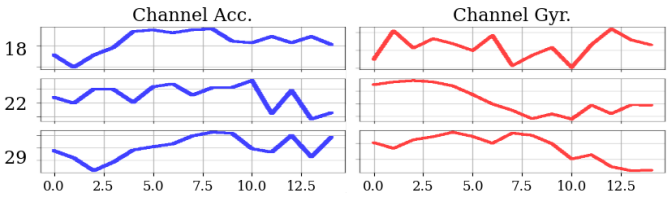


Fig. 13. Selected kernels from the first convolutional layer.

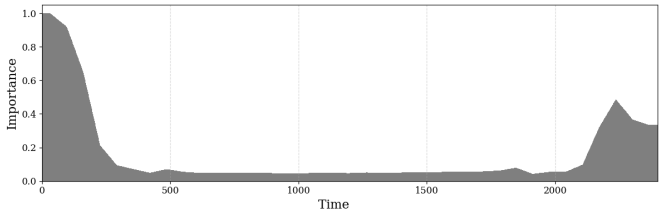


Fig. 14. Grad-CAM visualization of the importance of the input signal regions.

accompanying PDR-based displacement vectors, the estimated position at t_F can be rolled back in time to t_M, allowing comparison of the calculated position and heading with the absolute crosswalks on the map. A current limitation of this approach would be the inability to determine which crosswalk to associate with the detection. In areas with multiple crosswalks, we hypothesize that heading information will serve as a discriminative feature for resolving this ambiguity.

In this work, we do not assume a fixed device orientation, nor do we perform any orientation-based computations. Instead, we rely solely on signal magnitudes, reducing the available degrees of freedom but better reflecting real-world conditions, where smartphones are often carried in pockets or body-attached bags with variable orientations. Although using six degrees of freedom or gravity-based alignment could enhance training by isolating dominant acceleration vectors, we find that magnitudes alone provide a sufficiently robust representation.

VI. Conclusion

In this work, we detect crosswalk-crossings performed by PVI through their inertial measurements, with a motivation to use these detections for correction in inertial-based positioning algorithms. Aligned with this motivation, the main focus is to reach a biased accuracy in which false detections are not tolerable. Therefore, by imposing specific values for training parameters, we generate robust DCNN models that reach the required levels of detection performances in the test sets.

As the crosswalk-crossings are naturally rare events in normal walking conditions, we hit the class imbalance problem, which requires high amounts of data and time to train efficient models. We observe that undersampling the majority class at different levels can still generate reliable detection models consistent with our motivation, but with lower training times. We also show that the trained models are robust to different walking styles and environmental differences. Furthermore, we show that the models trained on the data collected with one device perform as robustly on the data of another device with different sensors, orientations and frequencies.

The current data are limited and include only a small number of PVI with varying degrees of impairment. This limitation can be addressed by enabling large-scale, community-driven collection of diverse inertial data and ground-truth labels via an accessible smartphone application designed for PVI. Moreover, we also observe that trained models perform similarly on the inertial data captured from different devices. The datasets can be enriched with different types of devices, and, more importantly, pretrained generic models can be fine-tuned to specialize for a specific device or individual. However, increasing the volume of the data brings further imbalanced set problems. To tackle this, the majority class can be undersampled more intelligently into smaller subsets, while maintaining instance diversity, which requires an appropriate metric to quantify instance similarity.

We finally investigate which parts of the input data and which convolutional features have impacts on the final decision. It is apparent that the beginning of a the input signal plays a significant role in detecting the crosswalk-crossings with a less but still significant role in the ending part. More importantly some of the convolutional features shine out to be really important in the decision process. As a future work, this analysis can be beneficial to redesign the input of the models, which would mitigate the time required for training and improve the quality of the models.

Acknowledgment

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